

PYTHON + NUMPY PROGRAMS

Q1: Install NumPy

```
!pip install numpy
```

Q2: Create a 2D NumPy array and print it

```
import numpy as np
arr = np.array([[1,2,3,4],
               [5,6,7,8],
               [9,10,11,12]])
print(arr)
```

Q3: Find item size of each element

```
import numpy as np
a = np.array([[1,2,3]])
print("Each item contains", a.itemsize, "bytes")
```

Q4: Find data type of array elements

```
import numpy as np
a = np.array([[1,2,3]])
print("Each item is of the type", a.dtype)
```

Q5: Check datatype with mixed values

```
import numpy as np
a = np.array([[1,'A',3]])
print("Each item is of the type", a.dtype)
```

Q6: Find size and shape of array (single row)

```
import numpy as np
a = np.array([[1,2,3,4,5,6,7]])
print("Array Size:", a.size)
print("Shape:", a.shape)
```

Q7: Find size and shape of array (two rows)

```
import numpy as np
a = np.array([[1,2,3,4],
             [8,5,6,7]])
print("Array Size:", a.size)
print("Shape:", a.shape)
```

Q8: Reshape an array

```
import numpy as np
a = np.array([[1,2],
              [3,4],
              [5,6]])
print("Original Array:")
print(a)
a = a.reshape(2,3)
print("Reshaped Array:")
print(a)
```

Q9: Find max, min and sum

```
import numpy as np
a = np.array([1,2,3,10,15,4])
print("Maximum value:", a.max())
print("Minimum value:", a.min())
print("Sum of elements:", a.sum())
```

Q10: Find square root and standard deviation

```
import numpy as np
a = np.array([[1,2,30],
              [10,15,4]])
print(np.sqrt(a))
print(np.std(a))
```

Q11: Arithmetic operations on arrays

```
import numpy as np
a = np.array([[1,2,30],
              [10,15,4]])
b = np.array([[1,2,3],
              [12,19,29]])
print(a + b)
print(a * b)
print(a / b)
```

Q12: Vertical and Horizontal stacking

```
import numpy as np
a = np.array([[1,2,30],
              [10,15,4]])
b = np.array([[1,2,3],
              [12,19,29]])
```

```
print(np.vstack((a,b)))  
print(np.hstack((a,b)))
```

Q13: Create NumPy array from list

```
import numpy as np  
l = [1,2,3,4,5,6,7]  
a = np.asarray(l)  
print(a)
```

Q14: Create NumPy array from tuple

```
import numpy as np  
t = (1,2,3,4,5,6,7)  
a = np.asarray(t)  
print(a)
```

Q15: Create NumPy array from multiple lists

```
import numpy as np  
l = [[1,2,3],  
     [8,9,10]]  
a = np.asarray(l)  
print(a)
```

Q16: Broadcasting example

```
import numpy as np  
a = np.array([[1,2,3,4],  
              [2,4,5,6],  
              [10,20,39,3]])  
b = np.array([2,4,6,8])  
print(a + b)
```

Q17: numpy.char.add()

```
import numpy as np  
print(np.char.add(['welcome','Hi'],  
                  [' to CRYSTAL',' read python']))
```

Q18: numpy.char.multiply()

```
import numpy as np  
print(np.char.multiply("hello ",3))
```

Q19: numpy.char.center()

```
import numpy as np  
print(np.char.center("CRYSTAL",20,'*'))
```

SHREYAS RAJU S (shreyasraju01@gmail.com)

GLOWLOGICS INTERNSHIP

ASSIGNMENT – 07

Q20: `numpy.char.capitalize()`

```
import numpy as np
print(np.char.capitalize("welcome to crystal"))
```

Q21: `numpy.char.title()`

```
import numpy as np
print(np.char.title("welcome to crystal"))
```

Q22: `numpy.char.lower()`

```
import numpy as np
print(np.char.lower("WELCOME TO CRYSTAL"))
```

Q23: `numpy.char.upper()`

```
import numpy as np
print(np.char.upper("welcome to crystal"))
```

```
[1]: !pip install numpy

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: numpy in .\AppData\Roaming\Python\Python313\site-packages (2.4.2)

[2]: import numpy as np

arr = np.array([[1,2,3,4],
               [5,6,7,8],
               [9,10,11,12]])

print(arr)

[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]

[3]: import numpy as np

a = np.array([[1, 2, 3]])
print("Each item contains", a.itemsize, "bytes")

Each item contains 8 bytes

[4]: import numpy as np

a = np.array([[1, 2, 3]])
print("Each item is of the type", a.dtype)

Each item is of the type int64

[5]: import numpy as np

a = np.array([[1, 'A', 3]])
print("Each item is of the type", a.dtype)

Each item is of the type <U21
```

SHREYAS RAJU S (shreyasraju01@gmail.com)

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```
[6]: import numpy as np

a = np.array([[1,2,3,4,5,6,7]])
print("Array Size:", a.size)
print("Shape:", a.shape)

Array Size: 7
Shape: (1, 7)
```

```
[7]: import numpy as np

a = np.array([[1,2,3,4],
              [8,5,6,7]])

print("Array Size:", a.size)
print("Shape:", a.shape)

Array Size: 8
Shape: (2, 4)
```

```
[8]: import numpy as np

a = np.array([[1,2],
              [3,4],
              [5,6]])

print("Original Array:")
print(a)

a = a.reshape(2,3)

print("Reshaped Array:")
print(a)

Original Array:
[[1 2]
 [3 4]
 [5 6]]
Reshaped Array:
```

```
Original Array:
[[1 2]
 [3 4]
 [5 6]]
Reshaped Array:
[[1 2 3]
 [4 5 6]]
```

```
[9]: import numpy as np

a = np.array([1, 2, 3, 10, 15, 4])

print("Maximum value:", a.max())
print("Minimum value:", a.min())
print("Sum of elements:", a.sum())

Maximum value: 15
Minimum value: 1
Sum of elements: 35
```

```
[10]: import numpy as np

a = np.array([[1, 2, 30],
              [10, 15, 4]])

print("Square Root:")
print(np.sqrt(a))

print("Standard Deviation:")
print(np.std(a))

Square Root:
[[1.         1.41421356  5.47722558]
 [3.16227766  3.87298335  2.        ]]
Standard Deviation:
10.044346115546242
```

SHREYAS RAJU S (shreyasraju01@gmail.com)

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ASSIGNMENT – 07

```
[11]: import numpy as np

a = np.array([[1, 2, 30],
              [10, 15, 4]])

b = np.array([[1, 2, 3],
              [12, 19, 29]])

print("Addition:")
print(a + b)

print("Multiplication:")
print(a * b)

print("Division:")
print(a / b)

Addition:
[[ 2  4 33]
 [22 34 33]]
Multiplication:
[[ 1  4 90]
 [120 285 116]]
Division:
[[ 1.         1.         10.        ]
 [ 0.83333333  0.78947368  0.13793103]]
```

```
[12]: import numpy as np

a = np.array([[1, 2, 30],
              [10, 15, 4]])

b = np.array([[1, 2, 3],
              [12, 19, 29]])

print("Vertical Stack:")
print(np.vstack((a, b)))
```

```
[12]: import numpy as np

a = np.array([[1, 2, 30],
              [10, 15, 4]])

b = np.array([[1, 2, 3],
              [12, 19, 29]])

print("Vertical Stack:")
print(np.vstack((a, b)))

print("Horizontal Stack:")
print(np.hstack((a, b)))

Vertical Stack:
[[ 1  2 30]
 [10 15  4]
 [ 1  2  3]
 [12 19 29]]
Horizontal Stack:
[[ 1  2 30  1  2  3]
 [10 15  4 12 19 29]]
```

```
[13]: import numpy as np

l = [1, 2, 3, 4, 5, 6, 7]
a = np.asarray(l)

print(a)

[1 2 3 4 5 6 7]
```

```
[14]: import numpy as np

t = (1, 2, 3, 4, 5, 6, 7)
a = np.asarray(t)

print(a)
```

SHREYAS RAJU S (shreyasraju01@gmail.com)

GLOWLOGICS INTERNSHIP

ASSIGNMENT – 07

```
[14]: import numpy as np

t = (1, 2, 3, 4, 5, 6, 7)
a = np.asarray(t)

print(a)

[1 2 3 4 5 6 7]
```

```
[16]: import numpy as np

l = [[1, 2, 3],
     [8, 9, 10]]

a = np.asarray(l)

print(a)

[[ 1  2  3]
 [ 8  9 10]]
```

```
[17]: import numpy as np

a = np.array([[1,2,3,4],
             [2,4,5,6],
             [10,20,30,3]])

b = np.array([2,4,6,8])

result = a + b

print(result)

[[ 3  6  9 12]
 [ 4  8 11 14]
 [12 24 45 11]]
```

```
[18]: import numpy as np

print(np.char.add(['welcome','Hi'],
                  [' to CRYSTAL',' read python']))

['welcome to CRYSTAL' 'Hi read python']
```

```
[19]: import numpy as np

print(np.char.multiply("hello ", 3))

hello hello hello
```

```
[20]: import numpy as np

print(np.char.center("CRYSTAL", 20, '*'))

*****CRYSTAL*****
```

```
[21]: import numpy as np

print(np.char.capitalize("welcome to crystal"))

Welcome to crystal
```

```
[22]: import numpy as np

print(np.char.title("welcome to crystal"))

Welcome To Crystal
```

```
[23]: import numpy as np

print(np.char.lower("WELCOME TO CRYSTAL"))

welcome to crystal
```

```
[24]: import numpy as np

print(np.char.upper("welcome to crystal"))
```